The Library MANAGEMENT SYSTEM

A Java-Based Object-Oriented Application

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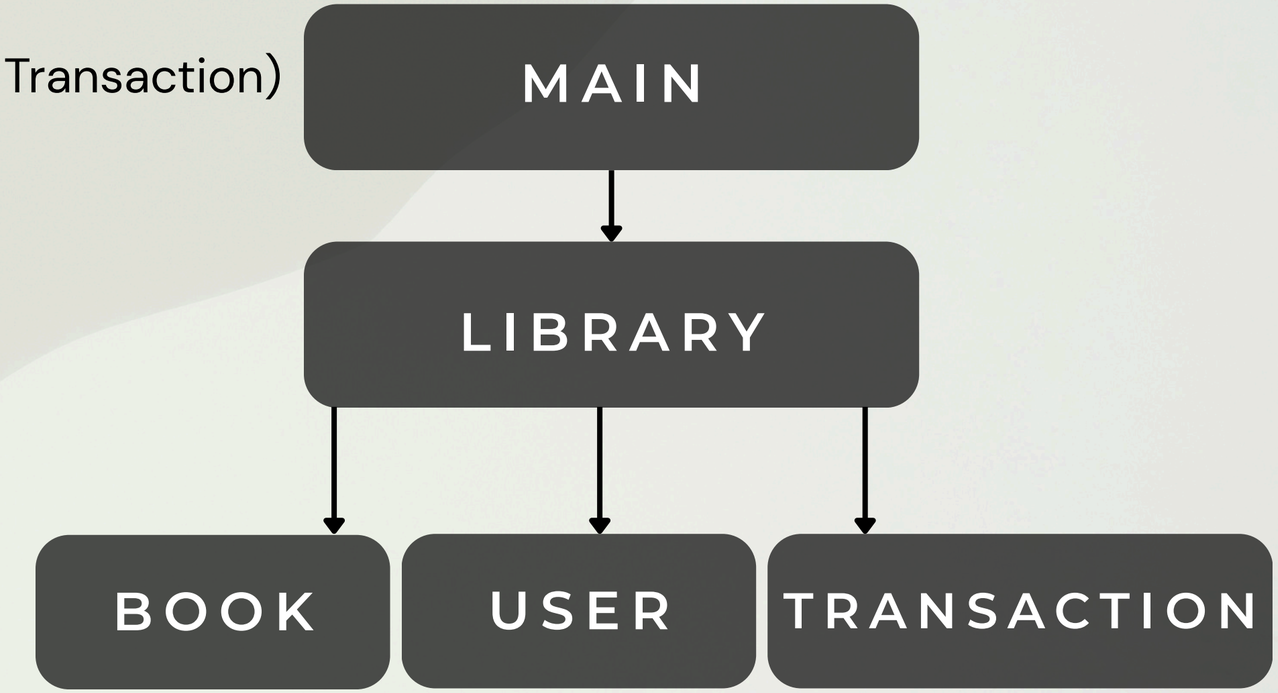
Target Users:
Students and Librarians in educational institutions

System Architecture

Architecture:
Main (CLI) → Library (Logic) → Models (Book, User, Transaction)

Key Classes:
Library – handles logic
User – abstract base class
Book – stores book data
Transaction – tracks operations

Package:
com.library (modular structure)



OOP Concepts

Abstraction

>Implemented using abstract User class defining common behavior

Inheritance

>Student and Librarian classes extend User for role specialization

Polymorphism

>getMaxBorrowLimit() overridden for different user types

Encapsulation

>Private fields with getters/setters ensure controlled data access

rary-java-app > src > com > library > Library.java

```java
public class Library {
    public User findUserById(String id) {
        for (User user : users) {
            if (user.getId().equalsIgnoreCase(id)) return user;
        }
        return null;
    }

    public boolean issueBook(String userId, String isbn) {
        User user = findUserById(userId);
        Book book = findBookByIsbn(isbn);

        if (user == null) {
            System.out.println("Error: User not found.");
            return false;
        }
        if (book == null) {
            System.out.println("Error: Book not found.");
            return false;
        }
        if (!book.isAvailable()) {
            System.out.println("Error: Book is already borrowed.");
            return false;
        }
        if (user.getBorrowedCount() >= user.getMaxBorrowLimit()) {
            System.out.println("Error: User has reached borrowing limit.");
            return false;
        }

        book.setAvailable(false);
        user.incrementBorrowed();
```

ROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Technical Highlight

Logic:
Validates availability and borrowing limits

Why:
Prevents invalid operations and ensures system consistency

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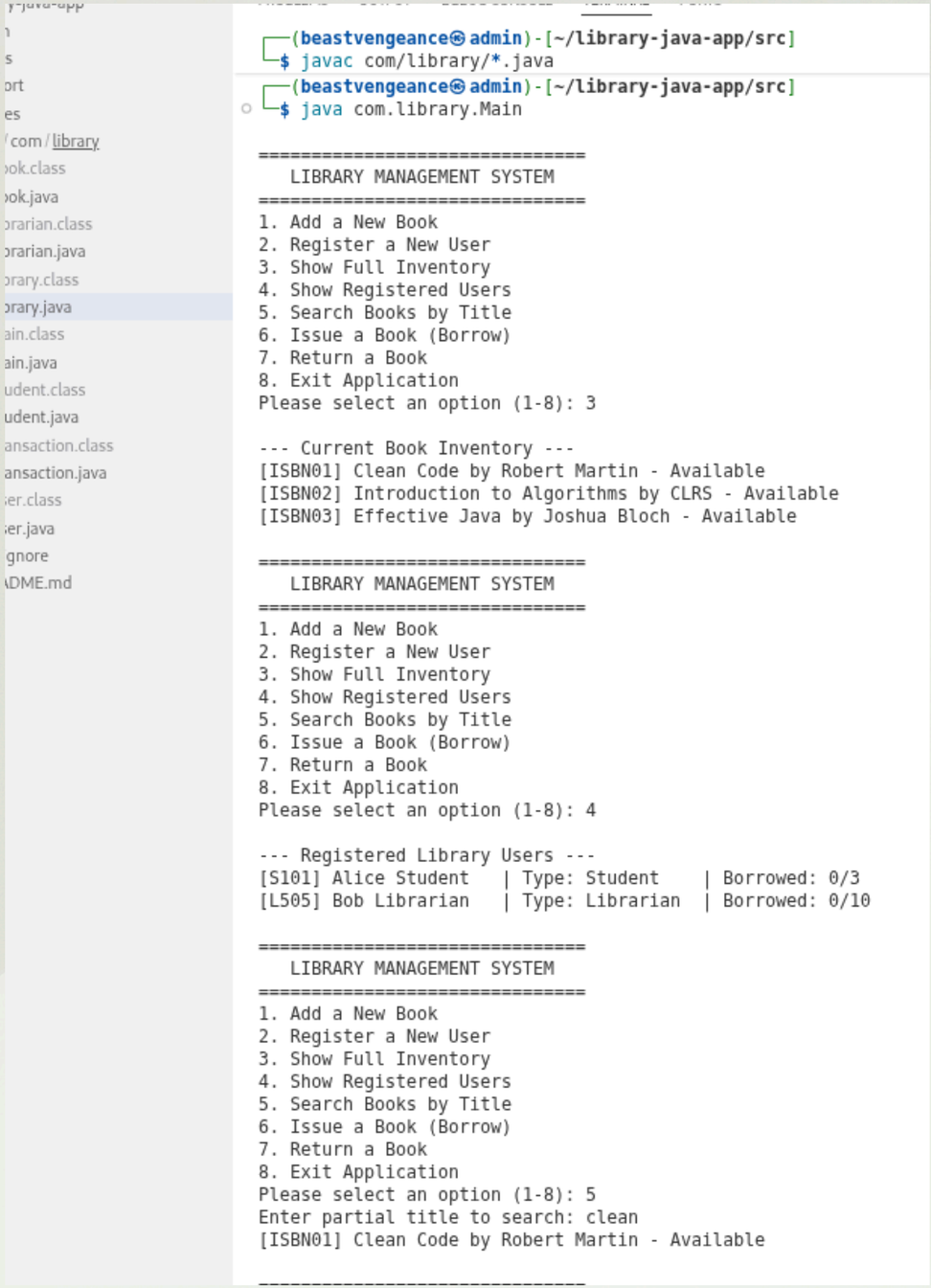
Live Demo


com / library
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DME.md
(beastvengeance㉿admin)-[~/library-java-app/src]
$ javac com/library/*.java
(beastvengeance㉿admin)-[~/library-java-app/src]
$ java com.library.Main
==========
LIBRARY MANAGEMENT SYSTEM
==========
1. Add a New Book
2. Register a New User
3. Show Full Inventory
4. Show Registered Users
5. Search Books by Title
6. Issue a Book (Borrow)
7. Return a Book
8. Exit Application
Please select an option (1-8): 3
--- Current Book Inventory ---
[ISBN01] Clean Code by Robert Martin - Available
[ISBN02] Introduction to Algorithms by CLRS - Available
[ISBN03] Effective Java by Joshua Bloch - Available
==========
LIBRARY MANAGEMENT SYSTEM
==========
1. Add a New Book
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7. Return a Book
8. Exit Application
Please select an option (1-8): 4
--- Registered Library Users ---
[S101] Alice Student | Type: Student | Borrowed: 0/3
[L505] Bob Librarian | Type: Librarian | Borrowed: 0/10
==========
LIBRARY MANAGEMENT SYSTEM
==========
1. Add a New Book
2. Register a New User
3. Show Full Inventory
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5. Search Books by Title
6. Issue a Book (Borrow)
7. Return a Book
8. Exit Application
Please select an option (1-8): 5
Enter partial title to search: clean
[ISBN01] Clean Code by Robert Martin - Available